MODELING DEMOCRACY NOTES

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1. Voting Systems

A voting system takes as an input the ranked preferences of a population on some finite number of options and outputs a collective choice (either of a subset of those options or a ranking of them). Here are some examples[1]:

First, we have voting rules which only take into account the pairwise comparison graph (PWCG), a directed graph on the set of candidates with weights of edges corresponding to the margins of match-ups between any pair of candidates in isolation.

PWCG-Based Rules:

- *PWC*: Whoever wins most head-to-head match-ups wins.
- *Sequential*: A tournament structure is fixed ahead of time. In each match-up, the winner is determined by the ranked-choice ballots. The winner of the tournament is selected.
- *Smith*: The winner is selected (using some other procedure) from within the Smith Set; that is, the smallest subset of candidates S such that no candidate outside S beats any candidate in S.
- *Beatpath (Schulze)*:
  - A *beatpath* is a directed path in the PWCG.
  - The *strength* of a beatpath is the least margin of any link in the path.
  - A candidate A *dominates* B if the strongest beatpath from A to B is stronger than the strongest beatpath from B to A.
  - In the Beatpath voting rule, the winner is selected (somehow) from among the candidates which are not dominated by any other candidate.
- *Ranked Pairs (Tideman)*: Edges in the PWCG are sequentially "activated" in order of greatest margin first, skipping any that would produce a directed cycle. The resulting graph will have a Condorcet candidate within that graph, and that candidate is selected.

Next, we have methods which repeatedly eliminate candidates using a runoff system:

Runoff-Based Rules:

- *Top-Two*: The top two in terms of first-choice votes proceed to a second runoff election. This can be determined by the relative ranking of the top two in the profile. Or alternatively, in the real world there can be a second election later, allowing time for more campaigning.
- *IRV (Instant-Runoff Voting)*: The candidates with least first-choice votes are repeatedly eliminated and their votes redistributed until one remains (only one election).
- *STV (Single Transferable Vote)*: In a situation where K winners must be selected, select a threshold T (often $V/(K+1)$ where V is the number of voters) and run the following algorithm until K candidates are elected:
  (1) If any candidate has more than T first-choice votes:
    - The one with the most votes, A, is elected.
    - The voters who elected A "spend" the portion of their remaining ballot corresponding to the portion that was necessary to reach T, so that the weight of their ballot for the remainder of the decision procedure is multiplied by $(M-T)/M$ where M is the (weighted) total of first-choice ballots for A.
    - A is removed from the pool of candidates and all ballots are consolidated.
    - Repeat step 1.

  (2) Otherwise, if no candidate has T first-choice votes, remove the candidate with the least first-choice votes. Then try step 1 again.
- *Coombs*: Like IRV except that the candidate with most last-choice votes is repeatedly eliminated.

[1]Note that many of these only identify sets of winners, within which tiebreaks must be conducted.

Next we have voting rules that are metric-based. A metric on ballots can be defined as follows: let the distance between two rankings be the least number of elementary moves required to get from one ballot to the other, where an elementary move consists of either swapping two adjacent-ranking candidates or (if ballots are permitted to be incomplete) adding/removing a candidate in last place. Then the following methods are possible:

Metric-Based Rules:

- *Dodgson*: From any profile, there is a least distance to an alternate profile where candidate A is Condorcet. Dodgson voting elects the candidate for which this distance is minimized.
- *Kemeny*: For every ordering of the candidates, we compute the distance to each ballot in the profile and sum these. The ordering in which this distance is minimized is selected, and a winner or multiple winners can be derived from this order.

Some rules privilege information about precise ranking levels rather than relative match-ups:

Position-Based Rules:

- *Plurality*: Whoever gets the most first-choice votes wins.
- *Borda*: Candidates are awarded a_1 points for each first place ranking, a_2 points for each second place ranking, etc. Typically $a_i = N - i$, where N is the number of candidates.
- *Condo-Borda (Black)*: If there is a Condorcet winner (i.e. option which wins all direct match-ups), it wins. Otherwise, proceed by Borda.
- *Secondality*: The candidate with the second-most first-choice votes wins (I think this is mostly a joke).
- *Plurality Veto*: Each candidate is awarded one point for each first-choice vote they receive. The voters are ordered in some way (which does effect the outcome) and each voter's ballot removes one point from their least-favored candidate remaining. When a candidate has lost all points, they are eliminated.

And finally we have the simplest of all voting rules:

Dictatorship:

- *Dictatorship*: One particular voter decides the result.

2. Criteria for Fairness

Voting systems can be judged based on various criteria for "fairness." Below is a list of these.

First, there are criteria based on guaranteeing sufficiently common ballot features are reflected in the outcome:

Consensus-Based Criteria:

- *Pareto Efficient*: If all ballots rank A first, then A must win.
- *Unanimity-fair*: If all ballots rank A above B, then the outcome must have A above B.
- *Majority-fair*: If the majority of ballots rank A first, then A must win.
- *Condorcet-fair*: If A is Condorcet then A must win.
- *Smith-fair*: Winners must come from the Smith set.

Next there are robustness criteria which restrict the effect that given changes to the ballots can have on the outcome:

Profile-Robustness Criteria:

- *Monotonic*: If A is winning and one ballot is altered in favor of A (so that A's ranking relative to all other candidates is no worse and all other relative rankings remain the same), then A still wins.
- *Strictly Monotonic*: If A is winning and one ballot is altered in favor or neutrally to A (so that A's ranking relative to all other candidates is no worse, and relative rankings unrelated to A may change), A still wins.
- *Strategy-Proof*: No voter can improve the outcome for themselves by voting any ranking other than their true preferences. Equivalently, altering a single ballot can only change the winner from A to B if it inverts their positions in favor of B[2].
- *Voter Anonymity*: The voting rule does not depend on the order of ballots.
- *Non-Dictatorship*: The outcome does not depend on only one voter.

Then there are criteria restricting the effect that given changes to the candidate pool can have on the outcome:

Candidate-Robustness Criteria:

- *No Spoilers*: If A is winning, removing B cannot cause A to lose, nor can adding in a candidate C, unless C is the sole winner. In these scenarios, B and C are called spoilers.
- *No Weak Spoilers*: There can be no spoilers that are weak, i.e. not in the Smith set.
- *Candidate Anonymity*: The voting rule does not depend on the order of candidates.

And finally, here is one quite strong criterion which requires robustness in both senses:

General Robustness Criteria:

- *Independence of Irrelevant Alternatives*: The relative outcome of candidates A and B cannot be affected by swapping any two candidates other than A and B on a given ballot *or* by adding or subtracting other candidates. In other words, whether $A < B$ in the outcome depends only on how each voter ranked A relative to B[3].
- *Modified IIA*: Whether $A < B$ in the outcome depends only on how each ballot ranked A relative to B *and* the distance between A and B (i.e. the number of candidates ranked between them) on each ballot.

[2]If this is not true, then voters may be incentivized to report their preferences falsely. For example, if ballot 1 and ballot 2 both have $A > B$, then a voter whose true preferences are ballot 2 should switch to ballot 1.

[3]If one assumes Voter Anonymity, IIA implies that whether $A \succ B$ depends only on the *number* of ballots preferring $A > B$ vs $B > A$, so the voting rule must be determined from the PWCG alone.

2.1. **Relationships Between Fairness Criteria.** To understand these criteria better, one should see how they relate, which are stronger and weaker, and so on. To begin with, here is a diagram of all the direct implications (a non-arrow indicates independence):

$$\begin{tikzcd}
 & \mathsf{SF} \arrow[d, Rightarrow] & & \\
 & \mathsf{CF} \arrow[d, Rightarrow] & & \mathsf{IIA} \arrow[d, Rightarrow] \\
\mathsf{UF} \arrow[dr, Rightarrow] & \mathsf{MF} \arrow[d, Rightarrow] & \mathsf{SM} \arrow[r, Leftrightarrow] \arrow[d, Rightarrow] & \mathsf{SP} \quad \mathsf{NS} \arrow[d, Rightarrow] \\
 & \mathsf{PE} & \mathsf{M} & \mathsf{NWS}
\end{tikzcd}$$

Most of these implications are trivial, with the one nontrivial fact being that SP and SM are equivalent; a proof of that is next. To show that there are no other implications:

- If the voting rule is *constant*, then it obeys both SM / SP and IIA (since no perturbation can change the outcome), but not PE. Thus, nothing on the middle or left can imply PE.
- **not done**

Claim: The following are equivalent:

(i) f is Strategy-Proof.
(ii) f is Strictly Monotonic.
(iii) f has the property that if an adjacent swap on a single ballot changes the outcome from A to B, then it must have been the swap $\binom{A}{B} \to \binom{B}{A}$.

Proof. (i) $\implies$ (ii): Suppose f is SP yet not SM, so there is some change to a ballot $b_1 \to b_2$ which is neutral/positive to A yet causes A to lose and B to win. Since f is SP, b_1 must prefer A to B (otherwise this change would be a strategic vote), and because the swap was neutral or positive to A, b_2 must also prefer A to B. But now the reverse change $b_2 \to b_1$ *is* a strategic vote for a voter whose true preferences are b_2, as it causes A to win, which b_2 prefers. This is a contradiction, so SP implies SM.

(ii) $\implies$ (iii): Suppose f is SM. This already implies that if an adjacent swap moves the winner from A to B, then it must *not* be neutral or positive for A, so in particular A must be swapped down. And looking in the reverse direction, which changes the winner from B to A, we can see similarly that the reverse swap must be negative for B. With these two facts, we can see that the swap must be $\binom{A}{B} \to \binom{B}{A}$.

(iii) $\implies$ (i): Suppose f has property (iii) but is not strategy-proof. Let $b_1 \to b_2$ be a ballot alteration which changes the winner from A to B and such that b_1 prefers B over A (a witness to the failure of SP). But I claim that I can change b_1 into b_2 through a sequence of swaps which will not allow B to win due to property (iii): divide the candidates into two sets, S_+ (those b_1 prefers to A) and S_- (A and those b_1 does not prefer to A). Then permute both sets so that each is ordered as in b_2. By (iii), the winner can only be in S_- now. Finally, elevate each candidate in S_- to its position in b_2. By (iii), none of these swaps can affect who the winner is. So finally, b_2 must have a winner in S_-, and in particular it cannot be B, which was assumed to be preferred to A by b_1, so f is Strategy-Proof. $\square$

There are a few more relations between combinations of these criteria:

- PE + SM $\implies$ UF: if every ballot has $A > B$ yet B is winning, then one can (neutrally to B) move A to the top of every ballot, which by PE makes A the winner but by SM shouldn't change the winner from B, a contradiction.
- M + IIA $\implies$ SM: IIA implies that if a swap changes the winner from A to B, then it must be between A and B, but does not forbid that the swap could be $\binom{B}{A} \to \binom{A}{B}$; M

removes this possibility because it is strictly positive for A, which forces the swap to be $\binom{A}{B} \to \binom{B}{A}$.

- **add more**

2.2. Voting Rules Compared by Fairness Criteria.

	PE	UF	MF	CF	SF	M	SM	NWS	NS
Plurality									
Borda									
Constant									
Dictator									
Beatpath									
Ranked Pairs									
IRV									
Coombs									
Dodgson									
Kemeny									

3. Impossibility Results

Having established some desirable fairness properties, we're forced to bite the bullet: many of them can never coexist in one voting rule.

<u>Incompatible Fairness Criteria</u>:

- *Majority Fair* and *No Spoilers*:

  Proof. Consider the following simple profile with a Condorcet cycle:

$\times 1$	$\times 1$	$\times 1$
A	B	C
B	C	A
C	A	B

  In principle, it should be difficult to name a winner here due to the symmetry. Indeed, by Majority fairness, A would lose if B were removed, B would lose if C were removed, and C would lose if A were removed. Thus if any of A, B, C were winning, they could be made to lose by removing a candidate, making the removed candidate a spoiler. So there are either spoilers or the system declares no winner. □

- (**Arrow**) $\mathsf{UF} + \mathsf{IIA} \implies$ Dictatorship:

  Proof. Call a set S of voters A-*decisive* if, whenever all of S ranks A first, A wins, and *decisive* if it is A-decisive for every candidate A. PE implies that the set of all voters is decisive, for instance. We can show using UF and IIA that for any decisive set S, if we break off one voter $v \in S$ to form $S = T \setminus \{v\}$, then either T is decisive or v is a dictator. By repeatedly doing this, we are guaranteed to find a dictator.

  Now suppose S is decisive, $v \in S$, and $T = S \setminus \{v\}$. Suppose that T is not decisive, so there is a profile for which T ranks A first, v does not, and some other candidate B wins. **not done** □

- (**Muller-Satterthwaite**) $\mathsf{PE} + \mathsf{SM} \implies$ Dictatorship:

  Proof. This only fails at 3 candidates or more (with 2, plurality works just fine).

  SM has some unreasonably strong consequences. Say a profile is on the (A, B)-*boundary* if A is currently winning and a single swap can make B win. Then SM implies that a profile can be on at most one boundary; if it were on both (A, X) and (A, Y), then there would be two swaps x, y on different ballots (different because SM implies that they are swaps $\binom{A}{X} \to \binom{X}{A}$ and $\binom{A}{Y} \to \binom{Y}{A}$ which cannot both be available on one ballot) that could change the outcome to X and to Y. But what if both are swapped? x changes the winner to X, and y is neutral to X, so the winner should be X, yet symmetrically it should be Y if they are performed in the other order—a contradiction.

$$\begin{tikzcd}
\boxed{A} & \begin{bmatrix} A & A \\ X & Y \end{bmatrix} \arrow[l, squiggly, "win"'] \arrow[r, "x"] \arrow[d, "y"'] & \begin{bmatrix} X & A \\ A & Y \end{bmatrix} \arrow[r, squiggly, "win"] \arrow[d, "y"] & \boxed{X} \\
\boxed{A} & \begin{bmatrix} A & Y \\ X & A \end{bmatrix} \arrow[l, squiggly, "win"'] \arrow[r, "x"'] & \begin{bmatrix} X & Y \\ A & A \end{bmatrix} \arrow[r, squiggly, "win"] & \boxed{X}
\end{tikzcd}$$

  Note also that the swap y keeps the profile on the (A, X)-boundary. More generally, if x changes the winner from A to X, then *any* swap y not between A and X in any other ballot must leave the profile in the (A, X)-boundary; otherwise, applying x after y must

result in A winning, so that applying y again switches the winner from A to X, implying that $B = X$ and $Y = A$ in the diagram below, i.e. y is $\binom{X}{A} \to \binom{A}{X}$.

$$\begin{tikzcd}
\boxed{A} & \begin{bmatrix} A & B \\ X & Y \end{bmatrix} \arrow[l, squiggly, "win"'] \arrow[r, "x"] \arrow[d, "y"'] & \begin{bmatrix} X & B \\ A & Y \end{bmatrix} \arrow[r, squiggly, "win"] \arrow[d, "y"] & \boxed{X} \\
\boxed{A} & \begin{bmatrix} A & Y \\ X & B \end{bmatrix} \arrow[l, squiggly, "win"'] \arrow[r, "x"'] & \begin{bmatrix} X & Y \\ A & B \end{bmatrix} \arrow[r, squiggly, "win"] & \boxed{A}
\end{tikzcd}$$

Now using these consequences of SM, we can establish dictatorship fairly easily: take a profile that is on the (A, B)-boundary for some A, B, and suppose that v is a ballot which has a pivotal swap available. Then without making any swaps of (A, B) one can elevate another candidate C to the top of every ballot other than v, leaving the profile still on the (A, B)-boundary. Then elevate C in v until it is between A and B. This pushes the profile out of the (A, B)-boundary, but the winner is still A because this is neutral to A. But now, C is unanimously preferred to B, **not done**

□

- (**Gibbard-Satterthwaite**) *Pareto Efficient*, *Strategy-Proof* and *Non-Dictatorship*:

  Proof. Equivalent to above, since $\mathsf{SM} \iff \mathsf{SP}$. □

4. Representation

In representative democracies, elected representatives of different districts are empowered to vote on their constituents' behalf. In some places, like the U.S. Senate, each of these districts (states) is given equal voting power as a way to intentionally increase the power of lower-population states. But in most cases, it is desirable to make some districts more powerful to account for their larger population or importance. There are two main ways to do this: either districts appoint multiple representatives (*apportionment*), with more being given to larger districts, or representatives of larger districts are given more weight in voting (*weighted voting*).

4.1. **Apportionment.** Suppose a population has size M, split among districts of populations $M_1, M_2, \ldots$, and the legislature has m seats. We'd like to use these population values to come up with a number of seats m_i for each district[4]. There are many different ways to do this:

Apportionment Methods:

(I) *Quota Methods*: A natural first idea is to define *quotas* $q_i := m \cdot (M_i/M)$. Ideally m_i should be near q_i, but m_i must be a natural number.
   - *Hamilton*: Begin with $m_i = \lfloor q_i \rfloor$. If there have not yet been m seats filled, add seats to districts in order of $q_i - m_i$.

(II) *Divisor Methods*: Another idea is to increase or decrease M until the quotas naturally add up to m.
   - *Jefferson*:
   - *Adams*:
   - *Webster*:
   - *Huntington-Hill*: [5]

But all of these methods have issues:

Apportionment Paradoxes:

- *Alabama Paradox*: When the total number of seats m is increased, and a district *loses* a seat as a result.
- *Population Paradox*: When one district's population share grows compared to another, yet it loses a seat to the other.
- *Quota Violation*: When a district ends up with a number of representatives that differs from its quota by more than 1.

Balinsky and Young showed that either population paradoxes or quota violations are inevitable in a deterministic apportionment method (though by using random chance we can get around this).

4.2. **Weighted Voting.** Suppose each district instead sends a single representative, and that representative is given voting weight w_i (normalized so that $\sum_i w_i = 1$). When representatives vote on a measure, we can say that it passes if the total weight of members voting for the measure is at least $\frac{1}{2}$ or some other threshold T. How should the weights be determined?

not done

[4]This was a big question in the early days of American democracy, hence the names of these methods

[5]This is what the U.S. house of representatives actually currently uses!

5. Modeling Elections and Voter Behavior

All of the previously discussed results assume that voters can rank candidates in any order, but in reality most rankings would be nonsensical, and few voters would actually vote that way. How likely is it that these paradoxical situations—spoiler effects, Condorcet cycles, etc.—actually arise in practice? One mechanism might be optimal in theory, but do worse in the distribution of real elections. We need a way to model elections with more fidelity to the real world.

One traditional model for voter preferences is within a *metric space*. If we assume that there is some finite number N of relevant continuous attributes (e.g. age, charisma, intelligence, support of a given social group, etc.) that a candidate can have, then we can model the space of candidates as $\mathbb{R}^N$ and place each candidate within that space accordingly. Then we can model a voter's preferences as a point in this space representing their ideal candidate, and their ranking can be obtained from the distances of the available candidates to this ideal.

Given a metric embedding into a space (X, d) of voters and candidates in a profile P, one can determine the *cost* of a candidate C as the total distance to C over all voters:

$$\text{cost}(C) := \sum_{v \in V} d(v, C).$$

It stands to reason that if a metric embedding truly represents the positions of voters and candidates in some sense, then the ideal candidate would be one which minimizes the cost. One could try to obtain metric information from voters, but self-reported data of this kind is highly unreliable (as well as being subject to strategic voting and so on). And rankings are not enough to determine who the lowest-cost candidate is.

Diagram

The natural next idea is to evaluate a voting rule f by worst-case cost (as a ratio of the optimal cost) over all possible profiles and embeddings; this is the *distortion* of f, denoted $\text{dist}(f)$:

$$\text{dist}(f) := \sup_{X,P} \frac{\text{cost}_X(f(P))}{\text{cost}_X(P_{optimal})}.$$

5.1. **Computing Distortion.** The distortion is typically difficult to compute, as the space of all metric spaces and all profiles is so large. But it is not too hard to at least give lower bounds by exhibiting particularly difficult profiles.

<u>Distortion Results:</u>

- $\text{dist}(f) \geq 3$ for all f.
  - To show this, consider a 2-voter 2-candidate profile in which the two voters have opposite preferences. Assume wlog that f selects A. It is possible to embed this profile into a metric space so that B is optimal and $\text{cost}(A)/\text{cost}(B)$ approaching 3:

    Diagram

    This can occur in any voting rule, so no voting rule can have distortion less than 3.
- If there is a Condorcet winner c, then $\text{cost}(c) \leq 3\,\text{cost}(a)$ for any other candidate a.
  - Because c is Condorcet, at least half of voters (say n) prefer c to a, and the other m prefer a to c. For voters among the n,

    $$d(v, c) \leq d(v, a)$$

    and for voters in the m, triangle inequality yields

    $$d(v, c) \leq d(v, a) + d(a, c).$$

    Combining these,

    $$\text{cost}(c) \leq \text{cost}(a) + m \cdot d(a, c).$$

    On the other hand, among the n which prefer c,

    $$d(v, a) \geq \tfrac{1}{2}\big(d(v, a) + d(v, c)\big) \geq \tfrac{1}{2} d(a, c)$$

which yields a lower bound on $\text{cost}(a)$ in terms of $d(a,c)$:

$$\text{cost}(a) \geq n \cdot \tfrac{1}{2} d(a,c) \implies 2\,\text{cost}(a) \geq n \cdot d(a,c) \geq m \cdot d(a,c).$$

Combining with the previous upper bound on $\text{cost}(c)$ gives

$$\text{cost}(c) \leq \text{cost}(a) + m \cdot d(a,c) \leq 3 \cdot \text{cost}(a).$$

– In general, if a portion β of voters prefer c to a, then

$$\frac{\text{cost}(c)}{\text{cost}(a)} \in \left[\frac{2}{\beta+1} - 1, \frac{2}{\beta} - 1\right].$$

- dist(Plurality) $= \infty$:
  – In a scenario with d candidates, consider a metric embedding in which one candidate A is at the origin and the other $d-1$ candidates are very near $(2,0)$. Distribute the voters with $\frac{1}{d} + \varepsilon$ at $(1-\varepsilon, 0)$ and split the remaining voters evenly at the positions of the other candidates.

  Diagram

  As $\varepsilon \to 0$, $\text{cost}(A)$ approaches $\frac{1}{d} + 2 \cdot (1 - \frac{1}{d})$ whereas the cost of any other candidate approaches $\frac{1}{d}$, giving a ratio of

$$\frac{\text{cost}(A)}{\text{cost}(B)} \to 2d - 1.$$

- dist(PluralityVeto) $= 3$:
  – Suppose each voter v vetoes candidate j_v, and let the overall winner be j^*. PV has the property that $d(v, j_v) \geq d(v, j^*)$. This immediately gives

$$\text{cost}(j^*) \leq \sum_v d(v, j_v) \leq \sum_v d(v,i) + d(i, j_v) = \text{cost}(i) + \sum_v d(i, j_v)$$

  for any other candidate i. **not done**
- dist(PluralityMatching) $= 3$:

A D3 voting rule minimizes distortion over all possible profiles, but for some profiles one can guarantee a better worst-case distortion (for example, if there is a Pareto winner then they are guaranteed to be metric-optimal, so minimality would require that the rule choose them). So let the condition *Minimal Distortion* mean that f minimizes worst-case distortion on every profile.

Question: For which profiles is the worst-case metric distortion less than 3?

- If C is Condorcet and receives at least β proportion of voters' support in all head-to-heads (so $\beta > \frac{1}{2}$) then C has distortion at most

$$\text{dist}(C) \leq \frac{2}{\beta} - 1 < 3.$$

  This does not imply that C is optimal. However, for every other candidate A, the worst-case distortion of A is at least

6. Project

Let the *biased metric* corresponding to the $\mathcal{C}$-indexed vector $(x_1, \ldots, x_m)$ be defined

define here

Ramakrishnan et al 2021 showed that within a given profile, the metrics with the greatest distortion for any candidate are all of this form.

Let A, B be two candidates in a profile P. We will use biased metrics to derive an upper bound on $\operatorname{cost}(A)/\operatorname{cost}(B)$ in embeddings where B is optimal. By Ramakrishnan 2021, we only need to consider biased metrics with $x_B = 0$, and we can normalize x_A to be 2. The biased metric d corresponding to $\vec{x}$ gives the following value for $\operatorname{cost}(A)/\operatorname{cost}(B)$:

$$\begin{aligned}\frac{\operatorname{cost}(A)}{\operatorname{cost}(B)} &= \frac{\sum_{v\in V} d(v, A)}{\sum_{v\in V} d(v, B)} \\ &= \frac{\sum_{v\in V} y_v + \min_{j:A\leq_v j} x_j}{\sum_{v\in V} y_v + \min_{j:B\leq_v j} x_j}\end{aligned}$$

Note that $\min_{j:B\geq_v j} x_j \leq x_B = 0$. Moreover, for voters who prefer A over B, $\min_{j:A\geq_v j} \leq x_B = 0$. Letting $[B > A] := \{v \in V : B \geq_v A\}$, we have the simplification

$$\begin{aligned}\frac{\operatorname{cost}(A)}{\operatorname{cost}(B)} &= \frac{\sum_{v\in V} y_v + \min_{j:A\leq_v j} x_j}{\sum_{v\in V} y_v + \min_{j:B\leq_v j} x_j} \\ &= \frac{\sum_V y_v + \sum_{[B>A]} \min_{j:A\geq_v j} x_j}{\sum_V y_v} \\ &= 1 + \frac{\sum_{[B>A]} \min_{j:A\geq_v j} x_j}{\sum_V y_v}.\end{aligned}$$

In producing an upper bound, it suffices to only consider biased metrics for which $0 \leq x_j \leq 2$ for all j; to see this, note that reducing each $x_j > 2$ to 2 can only decrease the denominator of the above fraction, and cannot affect the numerator. With that adjustment, $y_v = 1$ for all $v \in [A > B]$, so

$$\frac{\operatorname{cost}(A)}{\operatorname{cost}(B)} = 1 + \frac{\sum_{[B>A]} \min_{j:A\geq_v j} x_j}{\beta|V| + \sum_{[B>A]} y_v}$$

(where β is the proportion of voters who prefer A to B). This yields the interesting observation that the worst-case $\operatorname{cost}(A)/\operatorname{cost}(B)$ only depends on the ballots in $[B > A]$.

In the worst case, the highest $\operatorname{cost}(A)/\operatorname{cost}(B)$ can be is

$$\begin{aligned}\frac{\operatorname{cost}(A)}{\operatorname{cost}(B)} &= 1 + \frac{\sum_{[B>A]} \min_{j:A\geq_v j} x_j}{\beta|V| + \sum_{[B>A]} y_v} \\ &\leq 1 + \frac{\sum_{[B>A]} \min_{j:A\geq_v j} x_j}{\beta|V|} \\ &\leq 1 + \frac{\sum_{[B>A]} 2}{\beta|V|} \\ &= 1 + \frac{2(1-\beta)}{\beta} \\ &= \frac{2}{\beta} - 1.\end{aligned}$$

This agrees with the simple bound we arrived at first. This bound can be tight, though only for certain profiles. For this bound to be achieved, it must be that

$(*)$ For some (x_i), $y_v = 0 \ \forall v \in [B > A]$, or equivalently, $x_i \leq x_j \ \forall i \geq_v j, \ v \in [B, A]$.

And conversely $(*)$ forces $x_A \leq x_i$ whenever $A \geq_v i$ for $v \in [B > A]$, which implies that $\min_{j:A\geq_v j} x_j = 2$, showing that (x_i) achieves the bound. So $(*)$ is equivalent to $2/\beta - 1$ being tight.

Proposition: $(*)$ holds iff there is no chain of candidates

$$A = C_0 \geq_{v_1} C_1 \geq_{v_2} C_2 \geq_{v_3} \cdots \geq_{v_k} C_k = B$$

where $v_1, \ldots, v_k \in [B > A]$.

Proof. Define B^* to be the smallest set of candidates which contains B and is "upward closed" with respect to voters in $[B > A]$; that is, if $i \geq_v j$ for any $v \in [B > A]$ and $j \in B^*$ then $i \in B^*$. Similarly, define A_* to be the set of candidates generated downwards from A in the same manner.

$(*)$ implies that any $i \in B^*$ has $x_i = 0$, and any $j \in A_*$ has $x_j = 2$. In particular, $A_* \cap B^* = \emptyset$. This shows that no such chain can exist.

Conversely, if a chain does exist then at some point $x_{C_{i+1}} - x_{C_i}$ must be positive, because the sum of all of them is 2. This contradicts $(*)$. □

We can weaken $(*)$ to $(*\gamma)$, which says that excluding some subset S of voters with $|S|/|V| \leq \gamma$, no chain

$$A = C_0 \geq_{v_1} C_1 \geq_{v_2} C_2 \geq_{v_3} \cdots \geq_{v_k} C_k = B$$

exists for $v_1, \ldots, v_k \in [B > A] \setminus S$. If $(*\gamma)$ holds, then we can take a metric (x_i) where

$$\sum_{[B>A]} y_v \leq \gamma |V|, \quad \text{and} \quad \sum_{[B>A]} \min_{j : A \geq_v j} x_j \geq 2(1 - \beta - \gamma)$$

which shows that it is possible to achieve

$$\frac{\text{cost}(A)}{\text{cost}(B)} = 1 + \frac{\sum_{[B>A]} \min_{j : A \geq_v j} x_j}{\beta |V| + \sum_{[B>A]} y_v} \geq 1 + \frac{2(1 - \beta - \gamma)}{\beta + \gamma} = \frac{2}{\beta + \gamma} - 1.$$

One can think of $1 - \beta + \gamma$ as a proportion of voters who agree $B > A$ and have basically compatible preferences.

The more interesting question is how to best reconcile the preferences of voters who are incompatible.